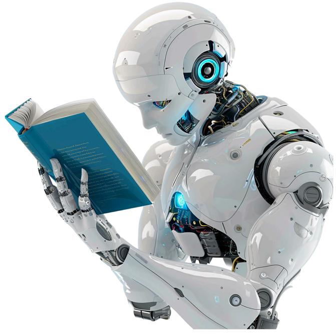

Midterm 1

General Instructions

In this midterm project, you will design, implement, and analyze **two independent machine learning pipelines** – one for **regression** and one for **classification** – using real-world datasets from Kaggle. **You must provide the original link where you downloaded the dataset from.**

You are free to choose any topic or application domain that interests you (healthcare, finance, sports, marketing, energy, environment, social data, etc.), as long as the chosen datasets and tasks satisfy the rules below.

This assignment is designed to evaluate your ability to:

- formulate meaningful ML problems,
- reason about modeling decisions,
- critically evaluate results, and
- clearly communicate your findings.

This is not a “follow-the-recipe” exercise. Thoughtful design, analytical depth, and interpretation will be rewarded.

If you are retaking this test, you must use different datasets. Failing to abide by this rule will result in 0 points.

Academic Integrity & Use of AI Tools – Mandatory Notice

This project must represent **your own independent work**.

Strictly prohibited:

- Submitting **AI-generated code** (including code produced by ChatGPT, Copilot, or similar tools).
- Copying code, notebooks, or solutions from **GitHub, Kaggle notebooks, blogs, or any other online sources**.
- Any form of **plagiarism, collaboration, or unauthorized assistance**.

Sanctions for violations:

Any violation of academic integrity will result in **severe disciplinary measures**, including:

- **Immediate annulment of all points obtained in this course so far**,
- **Failing the course**, and
- **Formal disciplinary procedures**, which may lead to **suspension or expulsion from the university**, in accordance with university regulations.

Code similarity detection and manual inspection will be applied.

If you use external sources **for learning or inspiration**, they must be **properly cited**, and all submitted code must be **written and fully understood by you**.

Dataset Rules

You must use **two different Kaggle datasets**:

- **One dataset for regression**
- **One dataset for classification**

For **each dataset**, you must:

- Provide a **direct link to the Kaggle page**.
- Use a dataset that:
    - Is **not built into scikit-learn**.
    - Contains **at least 1,000 samples (rows)**.
    - Has **at least 8 meaningful input features**.

You may:

- Clean the data.
- Perform feature engineering.
- Transform, encode, drop, and combine features.
- Handle missing values.

You may **not**:

- Use datasets included in scikit-learn.
- Use trivial or artificial datasets designed only for teaching.

Part I — Regression Task (13 points)

You will design and solve a **real-world regression problem** using:

- **Multiple Linear Regression**
  Regularized Regression (Ridge, Lasso, or ElasticNet)

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1. Problem Definition & Dataset Choice (3 pts)

- Clear problem formulation
- Meaningful choice of dataset
- Logical explanation of target variable

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2. Data Exploration & Preprocessing (4 pts)

- Exploratory data analysis
- Visualization and interpretation
- Proper preprocessing and feature engineering
- Handling missing values and outliers

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3. Modeling & Analytical Discussion (6 pts)

- Multiple linear regression baseline
- Multicollinearity analysis
- Regularization with at least two methods
- Model comparison
- Interpretation of coefficients
- Clear conclusions

Part II — Classification Task (13 points)

You will design and solve a **binary classification problem** using:

- **Logistic Regression**
- **k-Nearest Neighbors (k-NN)**
- **Decision Trees**

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1. Problem Definition & Dataset Choice (3 pts)

- Clear definition of classification task
- Meaningful problem formulation
- Justification of class labels

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2. Data Exploration & Preprocessing (4 pts)

- Exploratory data analysis
- Class balance analysis
- Visualization
- Feature preprocessing

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3. Modeling, Validation & Evaluation (6 pts)

- Logistic Regression, k-NN, Decision Tree
- K-fold cross-validation
- Hyperparameter tuning using GridSearchCV
- Proper use of:
    - Precision
    - Recall
    - ROC
    - AUC
- Model comparison and final selection

Final Comparative Analysis & Reflection (4 points)

You must submit a **separate written analysis** of **exactly one A4 page**, **font size 12**, addressing the following:

- Key challenges of both datasets
- Comparison of regression vs classification difficulties
- Algorithm strengths and weaknesses
- Bias–variance considerations
- Model limitations and failure cases
- What you would improve given:

    - More data
    - More time
    - More computational resources

This section evaluates your **critical thinking, clarity of expression, and analytical depth**.

Deliverables

1. Jupyter Notebook (.ipynb)

Must include:

- Clean, structured, readable code
- Clear markdown explanations
- Visualizations
- Logical organization

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2. Technical Report (PDF, 2–4 pages)

Must include:

- Problem formulation
- Key design decisions
- Interpretation of results
- Critical discussion
- Final conclusions